to align themselves in the same direction. They may not *all* align in the same direction, but they should at least cluster into domains that are all aligned. Let's get into the main training loop now that we have all of the necessary functions defined.

Listing 9.8. The 1D Ising model: The training loop

```
epochs = 200
lr = 0.001                                                    1
losses = [[] for i in range(size[0])]                         2
for i in range(epochs):
    for j in range(size[0]):                                  3
        l = j - 1 if j - 1 >= 0 else size[0]-1                4
        r = j + 1 if j + 1 < size[0] else 0                   5
        state_ = grid[[l,r]]                                  6
        state = joint_state(state_)                           7
        qvals = qfunc(state.float().detach(),params[j],
        layers=[(4,hid_layer),(hid_layer,2)])
        qmax = torch.argmax(qvals,dim=0).detach().item()      8
        action = int(qmax)
        grid_[j] = action                                     9
        reward = get_reward(state_.detach(),action)
        with torch.no_grad():                                 10
            target = qvals.clone()
            target[action] = reward
        loss = torch.sum(torch.pow(qvals - target,2))
        losses[j].append(loss.detach().numpy())
        loss.backward()
        with torch.no_grad():                                 11
            params[j] = params[j] - lr * params[j].grad
        params[j].requires_grad = True
    with torch.no_grad():                                     12
        grid.data = grid_.data
```

- ***1*** Learning rate
- ***2*** Since we're dealing with multiple agents, each controlled by a separate Q function, we have to keep track of multiple losses.
- ***3*** Iterates through each agent
- ***4*** Gets the left neighbor; if at the beginning, loops to the end
- ***5*** Gets the right neighbor; if at the end, loops to the beginning
- ***6*** state_ is the two binary digits representing the spins of the left and right neighbors.
- ***7*** state is a vector of two binary digits representing actions of two agents; turn this into
- ***8*** The policy is to take the action associated with the highest Q value.